



13. The time taken for a bacterial population to double is called:
 A) Growth curve B) Generation time
 C) Growth rate D) Reproduction speed
14. The type of fermentation observed in yeasts is
 A) acrylic fermentation B) lactic acid fermentation
 C) pyruvic fermentation D) alcoholic fermentation
15. Which of the following is a primary metabolite?
 A) Penicillin B) Amino Acid
 C) Toxin D) Pigment
16. What is the final electron acceptor in aerobic respiration?
 A) Oxygen B) Carbon dioxide
 C) Glucose D) ATP
17. Antibiotics are mainly produced by:
 A) Bacteria and fungi B) Viruses
 C) Protozoa D) Algae
18. Cheese is a product of -----
 A) Fermentation B) Distillation
 C) Pasteurisation D) Dehydration
19. Biofertilizers improve soil fertility by:
 A) Increasing temperature B) Killing pathogens
 C) Reducing soil pH D) Fixing nitrogen
20. What is the primary role of microbes in wastewater treatment?
 A) Removing heavy metals B) Producing antibiotics
 C) Increasing chemical pollutants D) Decomposing organic waste

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Give a comparative account of the structure and functions of prokaryotic and eukaryotic cells.	8	L*2	1	1,2
b) Summarize the general characteristics, classification, and reproduction of fungi.	8	L2	1	1,2
2. a) Describe the functions of bacterial flagella, pili, capsules, and other surface structures.	8	L2	1	1,2
b) Define sterilization and explain the operation of an autoclave, specifying the optimum conditions involved.	8	L1	2	1,2
3. a) Define pure culture and schematically represent Pour and spread plate method.	8	L1	2	1,2
b) Write short notes on: (i) Filtration methods of sterilization (ii) Negative staining	4+4	L1	2	1,2

Unit – II

4. a) Discuss the role of agar, and micronutrients in bacterial culture.	8	L2	3	1,7
b) Summarize the physical factors influencing microbial growth.	8	L2	3	1,7
5. a) Illustrate the bacterial growth curve and explain its different phases.	8	L2	3	1,7
b) Demonstrate with a schematic representation the mechanism of bacterial transformation with suitable examples.	8	L1	4	1,2,7
6. a) Define anabolism & catabolism and illustrate the citric acid cycle.	8	L1	4	1,2,7
b) Compare aerobic and anaerobic respiration in microorganisms.	8	L2	4	1,2,7

Unit – III

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| 7. | a) | Demonstrate the mechanism of phosphate solubilization by microorganisms and discuss its significance in agriculture. | 8 | L2 | 5 | 1,2, |
| | b) | Illustrate the microbial production of yoghurt and its health benefits. | 8 | L2 | 5 | 1,2, |
| 8. | a) | How is SCP produced on a large scale? Elaborate. | 8 | L2 | 5 | 1,2, |
| | b) | List out the applications of bioinsecticides in agriculture. | 8 | L2 | 5 | 1,2, |

Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
